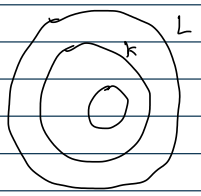
Lecture 7

Tuesday, 12 September 2023 10:29 AM

EXAFS & XANES } XAS

Structural info:
- + Oxidation state → mixture will be needed to sort out
- + Local geometry → XRD & planes; but this is photoelectrons (scattering electrons)
- + Surface coverage of Adsorbates
- + Valence orbital energies → This is where the chemistry happens

→ Build your own reactor to get in-situ data



K-shell (1s)

L (2s, 2p)

M (3s, 3p, 3d)

Energy of e^- it feels changes w/ the nucleus

↓

photons excite the e^- (1s, 2p e^- most likely → core e^-) } photoelectrons

↳ How do you make sure?

↳ You guess it based on your guess of what the element is & its concentration } You want to figure out if the metal is bonded & to what (oxidation states, etc..)

+ Can use for amorphous or crystalline materials

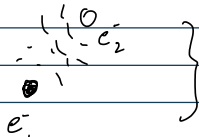
d-orbitals are doing the catalysis!

XAS:

Edge = peak

XANES → photo excitation (to unfilled orbital → via selection rules)

EXAFS → scattering (i.e. scattering due to greater than 1E e^- interacting w/ neighbors)



} Interference pattern

λ, distance, scattering efficiency

XPS → KE of photoelectron → records the e^- energy

XAS → records the photon energy (absorbed vs. not)

} Both give local structure!

Measure no. of photons in ion chamber (via charged plates)

$\log\left(\frac{I_0}{I_T}\right)$ ⇒ Absorbance

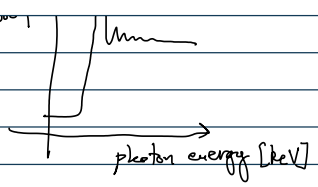
Flourescence (is just the inverse of absorbance)

Scattering is from neighbors!

Info:
- + No. of neighbors
- + Distance b/w atoms
- + Type of neighbor

Absorption

- Distance b/w atoms
- Type of neighbors
- Oxidation state



Diffraction needs long range order for interference patterns of x-rays

FT of Exafs →